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# MILITARY SPECIFICATION

## PROJECTOR, STILL PICTURE AP-4( )

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers one type of still picture projector capable of projecting 3.25 by 4-inch (82.6 by 101.6mm) slides designated as Projector, Still Picture AP-4( ) (see 6.4).

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| W-L-126 | - Lamp, Incandescent, Projection. |
| TT-P-1757 | - Primer Coating, Zinc Chromate, Low Moisture Sensitivity. |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Research and Development Command, ATTN: DELCS-PE, Fort Monmouth, NJ 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6730

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MILITARY

| | |
|---|---|
| MIL-P-116 | - Preservation, Method of. |
| MIL-C-675 | - Coating of Glass Optical Elements (Anti-Reflection). |
| MIL-C-3432 | - Cables, (Power And Special Purpose) And Wire, Electrical (300 And 600 Volts). |
| MIL-C-3767/13 | - Connector, Plug, Electrical (Power, Three-Wire, Fixed Grounding Type), Type UP221M. |
| MIL-P-11268 | - Parts, Materials, and Processes Used In Electronic Equipment. |
| MIL-F-14072 | - Finishes For Ground Electronic Equipment. |
| MIL-F-15160 | - Fuse; Instrument, Power, And Telephone. |
| MIL-E-55585 | - Electronics Equipment And Parts, Packaging Of. |

STANDARDS

FEDERAL

| | |
|---|---|
| FED-STD-H28 | - Screw Thread Standards For Federal Services. |
| FED-STD-595 | - Color. |

MILITARY

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures And Tables For Inspection By Attributes. |
| MIL-STD-130 | - Identification Marking of US Military Property. |
| MIL-STD-150 | - Photographic Lenses. |
| MIL-STD-454 | - Standard General Requirements For Electronic Equipment. |
| MIL-STD-726 | - Packaging Requirements Code. |
| MIL-STD-810 | - Environmental Test Methods. |

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issues listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE

| | |
|---|---|
| ANSI PH3.43 | - Projector Slides, Dimensions For. |
| ANSI S1.4 | - Sound Level Meters, Specification For. |

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)



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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. Projector, Still Picture AP-4( ) is a portable projector capable of projecting 3.25 by 4-inch (82.6 by 101.6mm) lantern slides. It consists essentially of a housing, a projection lens, a slide changer and associated accessories.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Parts, materials, and processes; general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirement therein, shall apply (see 4.4).

3.3.1 Cleaning. After assembly, the projector shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material.

3.3.2 Recycled, virgin and reclaimed materials. It is required that, to the maximum extent possible, recovered or reclaimed materials, in lieu of virgin materials, shall be used without jeopardizing the intended use of the manufactured item.

3.4 Design.

3.4.1 Wiring. All wires shall be of sufficient cross section to provide ample and safe current carrying capacity and mechanical strength. Where wires pass through holes in metal, they shall be protected by suitable grommets or bushings. Wires shall be as short as practicable except that sufficient slack shall be provided to prevent undue stress on wire and connections and to facilitate maintenance and repair of the projector. Wires shall be properly supported and secured to prevent undue change in position of the wires after the projector has been subjected to the conditions specified herein. Wires in a continuous run between two terminals shall not be spliced. The wiring layout shall be arranged to preclude exposure to detrimental high local temperatures such as that encountered in the vicinity of the projection lamp.

3.4.2 Fastening.

3.4.2.1 Screws, threads, and threaded parts. Screw threads shall conform to FED-STD-H28. The Unified Thread Form and Series shall be class 2A or 2B. American National Thread Series shall be class 2 fit. Threads in aluminum or other soft material shall not be used where frequent disassembly and reassembly may be necessary.



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3.4.2.2 Thread engagement and extensions. The length of thread engagement for tapped holes in steel or hard brass shall be not less than equal to the nominal thickness of corresponding machine screw nuts of size No. 12 and smaller and semifinished hexagon regular nuts in sizes .25 inch and larger as specified in FED-STD-H28. Thread engagement for tapped holes in aluminum, magnesium, or other soft material shall be not less than 50 percent greater than the thread specified for steel or hard brass. Where permitted by the construction of the projector, screws and bolts shall extend not less than 1.25 threads beyond the nut, but not more than 1.6 threads plus .125 inch (3.175mm) for screws not greater than one inch (25.4mm) in length and 1.5 threads plus .25 inch (6.35mm) for screws greater than one inch (25.4mm) in length.

3.4.3 Riveting. Rivets shall conform to MIL-STD-454, Requirement 12. Rivets shall be wet driven using a zinc-chromate primer conforming to TT-P-1757. For aluminum, the distance from the centers of the rivet holes to the edges of the material in which the rivets are placed shall be not less than twice the rivet diameter and 2.5 times for magnesium. The size and spacing of rivets on aluminum plates are sometimes dictated by factors other than those mentioned above. The following are some general rules that may be helpful in proportioning aluminum riveted joints and connections.

a. The diameter of a rivet should be not more than three times the thickness of the outside plates or shapes, and should be not less than the thickness of the thickest plate in which the rivet is driven.

b. A rivet used in the leg of an angle should have a diameter not more than 0.3 times the width of the leg.

c. The minimum spacing of rivets should be three times the nominal rivet diameter. The maximum spacing is often governed by the possiblity of compression buckling of parts between rivets. Where buckling or strength requirements do not control, the spacing should still be held below some reasonable maximum limit, such as 24 times the thickness of the sheet or plate.

d. In riveted joints that must be made pressure tight by caulking the edge of the plates, the edge distance should be not more than 1.5 times the diameter nor ten times the thickness of the plate.

3.4.4 Electrical input. The projector shall be capable of operating with the voltage ranging from 109 to 129 volts alternating current and at frequencies from 47.5 to 63.0 Hertz (Hz).

3.4.4.1 Overload protection. The power circuit of the projector ventilation motor shall be fused. The fuses shall be so located as to provide easy and unobstructed access and arranged to permit safe and easy replacement without removal of the projector housing. Fuses shall conform to MIL-F-15160. Both sides of the line shall be fused.



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3.4.4.2 Dielectric strength. There shall be no flashover or breakdown when potentials specified in 4.6.1 are applied between any electrical circuit, or between any electrical circuit and adjacent metal parts not in the circuit under test.

3.4.5 Interchangeability. Like units, assemblies, sub-assemblies and replaceable parts shall be physically and functionally interchangeable, without modification of such items or the projector. Individual items shall not be hand picked for fit or performance.

3.5 Durability. The projector shall operate continuously for not less than 100 hours with the slide changer operated for not less than 50,000 cycles without excessive wear, damage or malfunction.

3.6 Noise level. Noise level at a distance of three feet from the projector in a horizontal plane passing through the base of the projector shall be not more than 55 decibels above $10^{-16}$ watt/cm$^2$ under free-field conditions. If the noise spectrum is such that the peak is more than 575 Hz, the articulation index shall be not less than 0.4 when measured in accordance with 4.6.4.

3.7 Safety measures.

3.7.1 Mechanical. The projector shall be such that, when set up for operation, accidental contact by personnel with moving parts such as fans will be prevented. Projecting and overhanging edges shall be held to a minimum. Edges and corners of enclosures shall be rounded.

3.7.2 Electrical. Provisions shall be made to prevent personnel from accidentally coming into contact with electrically energized parts. The contractor shall submit to the procuring activity proof that the commodity furnished under this specification conforms to the Standards of Underwriters Laboratories, Incorporated, regarding fire and casualty hazards. A label, marker, listing, or certified report of the Underwriters Laboratories, Incorporated, will be accepted as evidence of compliance with this requirement. The complete projector shall be grounded through the power cord to the power receptacle.

3.8 Performance.

3.8.1 Light output. When tested as specified in 4.6.5.1, the light output of the projector shall be not less than 700 lumens.

3.8.2 Uniformity of screen illumination. Illumination on the screen shall be free from bands or patches differing in color or brightness from the adjacent parts of the screen. The average illumination at the four corners shall be not less than 75 percent of the illumination at the center. At no corner shall the illumination be less than 65 percent of the illumination at the center of the screen (see 4.6.5.2).

3.8.3 Distortion. When a test plate is projected as specified in 4.6.5.3, the projected image of the lines adjacent to both the long and short edges of the picture aperture shall depart not more than 0.5 percent of their length from a straight line joining opposite ends of the projected image of the lines.



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3.8.4 Resolution. Within the picture area centered with respect to the lens axis, the resolving power at the center of the picture area shall be not less than 40 lines/mm. The resolving power elsewhere shall be not less than 14 lines/mm (see 4.6.5.4).

3.8.5 Temperature rise.

3.8.5.1 Slide. The temperature rise, when measured as specified in 4.6.5.5.1, shall not exceed 135 Fahrenheit (57 Celsius) degrees above an ambient temperature of 70°F (21°C).

3.8.5.2 Projector. At the conclusion of two hours of continuous operation, no temperature rise more than 55 Fahrenheit (31 Celsius) degrees above an ambient temperature of 70°F (21°C), shall be encountered on any external part of the projector exclusive of the areas immediately above the lamp house and immediately adjacent to the blower exhaust port. At the conclusion of 8 continuous hours of operation, there shall be no damage to any part of the projector, including the projection lamp, due to excess heat (see 4.6.5.5.2).

3.9 Environmental.

3.9.1 Temperature.

a. Operating. Ambient temperature in the range of 32°F (0°C) to 110°F (43°C).

b. Storage. Exposure in the range of -60°F (-51°C) to 160°F (71°C). Exposure at the high temperature extreme not more than four hours and at the low temperature extreme not more than 24 hours, at any one time.

3.9.2 Relative humidity. Up to 97 percent relative humidity for 20 hours and exposure at 100 percent relative humidity, with condensation, for four hours.

3.9.3 Elevation.

a. Operating. Up to 10,000 feet (3048m) above sea level.

b. Nonoperating. Up to 50,000 feet (15,240m) above sea level.

3.9.4 Drop. When packed in the carrying case ready for hand carrying, the projector, except for installed projection lamps, shall suffer no physical damage by dropping.

3.9.5 Vibration. The projector shall be capable of withstanding vibration below 55 Hz without suffering damage due to mechanical resonance.

3.9.6 Fungus. When tested as specified in 4.6.6.7, the projector (including accessories) shall show no evidence of viable fungus or corrosion. Corrosion is any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.

3.10 Details of components.



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3.10.1 Body. The construction of the projector shall be such that, in the replacement of interchangeable parts, no adjustments are required that cannot be performed by a serviceman using standard tools. Each major subassembly, except the lens assembly, shall provide easy access to its interior parts, terminals, and wiring for complete circuit checking and for the removal and replacement of parts. Wires, cables, parts, or assemblies shall not be removed in order to gain access to terminals, soldered connections, mounting screws, etc. Where it is not practical to avoid such construction, those parts which must be displaced or removed shall be so constructed, mounted, and otherwise arranged as to facilitate their displacement or removal without being disconnected from their circuits.

3.10.1.1 Tilt device. Means shall be provided for tilting the projector from the horizontal to 8° above the horizontal. The tilt device shall be spring loaded and shall have an adjustment mechanism that will permit quick, easy, and exact adjustment of the projector to the required angle. When set, the adjustment shall remain fixed until changed by the operator.

3.10.1.2 Physical stability. The projector, set up for operation, shall not fall over when the normally horizontal supporting surface is tilted 20° from the horizontal in any direction with the tilt device fully extended.

3.10.1.3 Rigidity.

3.10.1.3.1 Slide changer. The average time for cessation of observable vibrations shall be not more than three seconds when the slide changer is operated.

3.10.1.3.2 Ventilation motor. Vibration caused by the ventilation motor shall not visibly affect the projected image when the tilt device is at full extension.

3.10.1.3.3 Projector. When tested as specified in 4.6.9.3, the average time for cessation of observable vibrations shall be not more than five seconds.

3.10.1.4 Ventilation. A motor and ventilating fan shall be provided as an integral part of the projector. The fan shall be affixed to the motor shaft. The motor shall be provided with a self-contained supply of lubricant sufficient for not less than 2,000 hours of continuous operation. The lubrication system shall be leakproof, spillproof, dripproof, and dustproof in any position.

3.10.1.5 Switches. Switches shall be provided for energizing the ventilation motor and the projection lamp successively. The circuits shall be so arranged that the projection lamp cannot be energized without first energizing the ventilation motor. In lieu of individual motor and lamp switches, one three-position rotary switch shall be acceptable with positions as follows:

| Switch position | Circuit energized |
|---|---|
| Left | Off |
| Center | Motor |
| Right | Motor and lamp |



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3.10.2 Optical system.

3.10.2.1 Lens. The projector shall be provided with an anastigmatic, type VII, as defined in MIL-STD-150, f/5.3, or faster, projection lens with an effective focal length of 12.5 inches (317.5mm) ±3 percent. All air-to-glass surfaces shall be coated with a reflection reducing coating in accordance with MIL-C-675. The lens shall meet the applicable requirements of MIL-STD-150 for type VII lenses.

3.10.2.2 Mounting. The lens mounting shall maintain the lens centered on the optical axis and its axis perpendicular to the lantern slide.

3.10.2.3 Focusing mechanism. The focusing mechanism shall be capable of producing satisfactory screen images for projected widths from 3 to 20 feet (.91 to 6.1m). A coarse and a fine focusing adjustment mechanism shall be provided. The fine focusing mechanism shall be rack and pinion gear, and shall prevent creeping or movement of the focusing mechanism during projector operation.

3.10.2.4 Bellows. The projector shall be furnished with a folding type bellows that will permit the lens mounting standard to be moved to the limits of travel afforded by the focusing mechanism. The bellows shall be light tight and shall be fabricated from a moisture-and-fungus resistant material. The bellows shall not sag so as to interfere with the projected image.

3.10.2.5 Condenser lenses. The projector shall be supplied with a set of condenser lenses in accordance with MIL-STD-150, type XI. The lenses shall be mounted in an assembly which shall be removable from the projector as a complete unit.

3.10.2.6 Heat retarding filter. A glass heat retarding filter shall be mounted in the condenser assembly.

3.10.2.7 Accessibility of optical parts for cleaning. The surfaces of the condenser lenses, heat retarding filter, and the front surface of the mirror used behind the lamp shall be accessible for cleaning without the use of tools. Lens and mirror mounts shall be so designed as to insure the accurate return of lenses and mirrors to their original positon when they are replaced after having been removed for cleaning.

3.10.2.8 Built-in pointer. The projector shall have a built-in pointer which can be adjusted so that the image will pinpoint any area of the projected picture.

3.10.3 Light source.

3.10.3.1 Projection lamp. The projection lamp shall be medium prefocus, base-down, conforming to W-L-126, ASA Code DDB 750 watt.

3.10.3.2 Lamp and reflector adjustment. A lamp and reflector adjustment shall be provided to permit the lamp and reflector to be positioned for maxium light output.



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3.10.3.3 Light shielding of lamp housing. The lamp housing and vents thereto shall be constructed to prevent excessive light spill from the projection lamp which would interfere with the operator or distract the viewer's attention from the material being projected.

3.11 Power Cable. The projector shall be furnished with a 15 +0.5 feet (4.6 +.15m) power cable. The cable shall conform to MIL-C-3432, type CO-03MGF(3/16)0365. One end of the cable shall be provided with an integrally molded female receptacle to fit a standard three-blade receptacle mounted flush on the projector. The other end shall be provided with a connector plug conforming to MIL-C-3767/13, type UP221M.

3.12 Slide changer. The slide changer shall accommodate standard 3.25 by 4-inch (82.6 by 101.6mm) lantern slides conforming to ANSI PH3.43. The construction shall be such that the preceding slide may be removed and the slide to follow inserted without disturbing the slide in the projection position. The slide changer shall have stops which shall automatically center the slide carrying pockets in position across the center of the optical system as each slide is positioned for projection. The slide changer shall be easily removable from and insertable into the projector without the use of tools.

3.13 Carrying case. A carrying case shall be provided to house and protect the projector and the following items from weather conditions and from blows and impact during transportation and storage:

a. Power cord (1 each).
b. Slide changer (1 each).
c. Projection lamp (2 each).
d. Fuse (2 each).
e. Lens tissue and brush.

Carrying handles shall be located in a horizontal position, 12 inches (304.8mm) up from the bottom on each end of the case. The construction of the case shall permit stacking of cases.

3.14 Dimensions and weight. The carrying case containing the projector and accesories shall be not more than 27.5 inches (698.5mm) long, 9.5 inches (241.3mm) wide and 15.5 inches (393.7mm) high and shall weigh not more than 50 pounds (22.7kg).

3.15 Finish, protective. All protective coatings and finishes shall conform to MIL-F-14072.

3.15.1 Final paint film. The final paint film on exterior surfaces, or any interior surfaces which are visible when the equipment is in use, shall be either lusterless green, conforming to FED-STD-595, color chip No. 34087, or lusterless black, conforming to FED-STD-595, color chip No. 37038. Interior surfaces, except those requiring a polished finish shall be lusterless black conforming to FED-STD-595, color chip No. 37038.

3.16 Identification marking. The projector and carrying case shall be identified in accordance with MIL-STD-130.



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3.17 Workmanship. The projector shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs herein:

3.3.1 Cleaning
3.4.1 Wiring
3.4.3 Riveting

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Inspection covered by subsidiary documents (see 4.4).
c. Quality conformance inspection (see 4.5).
d. Packaging inspection (see 4.7).

4.3 First article.

4.3.1 First article units. When required, and as specified in the contract, the contractor shall furnish first article units of the complete projector.

4.3.2 First article inspection. This inspection shall consist of the inspection specified in the subsidiary document covering the item listed in 4.4 and the inspection specified for group A, group B, and group C (see tables I, II, and III, respectively). Failure of any inspection shall be cause for rejection of the first article.

4.4 Inspection covered by subsidiary document. The finish, as specified in 3.15 shall be inspected under the applicable subsidiary document as part of the inspection requirements.

4.5 Quality conformance inspection.

4.5.1 Inspection. Quality conformance inspection shall be as specified in 4.4 and 4.5.2 through 4.5.4.2.

4.5.2 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in table I. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105, using the general inspection levels and AQLs indicated in table I.



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TABLE I. Group A inspection.

| Inspection | Requirement paragraph | Test method paragraph | AQL (percent defective) | |
|---|---|---|---|---|
| | | | Major | Minor |
| Visual and mechanical<br>Operation | 3.17<br>3.10.1.1,<br>3.10.2.3 | 4.6.10<br>4.6.7 | 1.5<br>1.0 | 6.5 |

4.5.2.1 Order of inspection within group A. Group A inspection shall be performed in any order satisfactory to the Government.

4.5.3 Group B inspection. Group B inspection shall be as specified in table II. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

TABLE II. Group B inspection.

| Inspection | Requirement paragraph | Test method paragraph | AQL (percent defective) |
|---|---|---|---|
| Interchangeability | 3.4.5 | 4.6.2 | 6.5 |
| Distortion | 3.8.3 | 4.6.5.3 | 4.0 for the combined group |
| Light output | 3.8.1 | 4.6.5.1 | |
| Resolution | 3.8.4 | 4.6.5.4 | |
| Uniformity of screen illumination | 3.8.2 | 4.6.5.2 | |

4.5.3.1 Group B sampling. Sampling for group B inspection shall be in accordance with MIL-STD-105, inspection level S-4. AQL shall be as specified in table II.

4.5.3.2 Order of inspection within group B. Group B inspection shall be performed in any order satisfactory to the Government.

4.5.4 Group C inspection. Group C inspection shall be as specified in table III and shall be performed on sample units that have been subjected to and passed groups A and B inspection.



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Table III. Group C inspection.

| Inspection | Requirement paragraph | Test method paragraph |
|---|---|---|
| Group I | | |
| Dielectric strength | 3.4.4.2 | 4.6.1 |
| Moisture resistance | 3.9.2 | 4.6.6.3 |
| Group II | | |
| Temperature | 3.9.1 | 4.6.6.1, 4.6.6.2 |
| Elevation | 3.9.3 | 4.6.6.4 |
| Drop | 3.9.4 | 4.6.6.5 |
| Vibration | 3.9.5 | 4.6.6.6 |
| Fungus | 3.9.6 | 4.6.6.7 |
| Group III | | |
| Noise | 3.6 | 4.6.4 |
| Temperature rise | 3.8.5 | 4.6.5.5 |
| Physical stability | 3.10.1.2 | 4.6.8 |
| Rigidity | 3.10.1.3 | 4.6.9 |
| Group IV | | |
| Durability | 3.5 | 4.6.3 |

4.5.4.1 Group C sampling. One sample unit shall be selected at random for each group of inspections in table III from every 200 projectors and fraction thereof produced, or every three months' production, whichever occurs first. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot.

4.5.4.2 Group C failures. Action required relative to group C failures shall be as specified (see 6.2).

4.6 Methods of inspection.

4.6.1 Dielectric strength. The insulation strength of all electrical circuits shall be tested to withstand the rated voltage plus 1000 volts. The test potential may be alternating current or direct current, of the same value, and shall be applied for one



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minute between all wiring and ground. This test shall be performed immediately following the humidity test. Nonconformance to 3.4.4.2 shall constitute failure of this test.

4.6.2 Inspection for dimensional interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability specified in 3.4.5. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, housing, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, housings, and mountings.

d. Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts.

e. Size and form of special threads.

4.6.3 Durability. The projector motor and lamp shall be energized for 100 continuous hours. During this time, the slide changer shall be operated for 50,000 cycles. The slide changer shall be operated at a speed not more than two frames per second. This test shall be interrupted only to replace burned out projection lamps. At the completion of this test, the component parts of the projector shall be examined. Any evidence of excessive wear, damage or malfunction shall constitute failure of this test.

4.6.4 Noise. The projector shall be operated in a location where free-field conditions prevail and the noise spectrum shall be measured with a noise spectrum analyzer. Measurements of the sound spectrum shall be made using a type 9700 analyzer manufactured by GENRAD, Inc., Concord, MA, or a Model 3580A analyzer manufactured by Hewlett-Packard Co., Palo Alto, CA, or equal and a sound level meter conforming to ANSI S1.4. The sound level meter shall be set at the curve "C" setting when analyzing sound levels and frequency, and at the curve "B" setting when measuring sound levels only. If the peak of the noise spectrum is 575 Hz or less, measure the noise level at 45° intervals around a circle in a horizontal plane passing through the base of the projector and at a distance of 3 feet (91.4cm) from the projector. If the peak of the noise spectrum is above 575 Hz, measure the noise level in each of the following 20 bands.

| | | | |
|---|---|---|---|
| a. 250-375 | f. 955-1130 | k. 1930-2140 | p. 3255-3680 |
| b. 375-505 | g. 1130-1315 | l. 2140-2355 | q. 3680-4200 |
| c. 505-645 | h. 1315-1515 | m. 2355-2600 | r. 4200-4860 |
| d. 645-795 | i. 1515-1720 | n. 2600-2900 | s. 4860-5700 |
| e. 795-955 | j. 1720-1930 | o. 2900-3255 | t. 5700-7000 |



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In making the measurements take three readings 3 feet (91.4cm) from the projector in the manner previously described. Compute the average value for each frequency band. Using the following formula compute the articulation index:

$$A = 0.896 - 0.00167 (B_1 + B_2 + B_3 + .... B_{20})$$

Where:

A= Articulation index.

B= Long average intensity per cycle of the noise level in db above $10^{-16}$ watt/$cm^2$ for each band.

Note: Long average intensity is equal to one half the average observed intensity in each band.

4.6.4.1 Failure criteria. Nonconformance to 3.6 shall constitute failure of this test.

4.6.5 Performance.

4.6.5.1 Light output. The projector shall be set up for operation with a 3.25 by 4-inch (82.6 by 101.6mm) slide projected on a screen located in a plane perpendicular to the optical axis of the projector. The width of the image shall be 60 inches (1524mm). The slide shall then be removed and the illuminated area of the screen shall be divided into 25 squares. The incident illumination at the center of each square shall be measured by means of a photo light meter corrected by suitable filters to conform within a good approximation to the visibility curve of the eye. The mathematical average of the 25 illumination measurements in foot-candles shall be multiplied by the area of the illuminated screen in square feet to obtain the light output in lumens. Nonconformance to 3.8.1 shall constitute failure of this test.

4.6.5.2 Uniformity of screen illumination. The projector shall be set up for operation with a 3.25 by 4-inch (82.6 by 101.6mm) slide projected on a screen to produce a 60-inch (1524mm) wide image. The screen shall be located in a plane perpendicular to the optical axis of the projector. The screen illumination shall be measured at the center and four corners after the slide has been removed. The corner measurements shall be made at 3 inches (76.2mm) from top or bottom edge and 3 inches (76.2mm) from either side of the projected image. Nonconformance to 3.8.2 shall constitute failure of this test.

4.6.5.3 Distortion. A test plate having straight lines adjacent to both the long and short edges of the picture aperture shall be projected on a white, matte, grainless screen located at such distance from the projector that the horizontal dimension of the projected image is 40 inches (1016mm). Straight lines shall be drawn on the screen connecting the end points of the projected lines. (The border lines of the resolving power test chart may be used for the test plate.) Nonconformance to 3.8.3 shall constitute failure of this test.



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4.6.5.4 Resolution. The resolving power of the lens shall be tested in accordance with MIL-STD-150, Method 15. Nonconformance to 3.8.4 shall constitute failure of this test.

4.6.5.5 Temperature rise.

4.6.5.5.1 Slide. A thermocouple of 0.00396 or 0.00353 inch (.1006 or .0897mm) diameter (38 or 39 AWG) wire shall be cemented or held in contact at the center of a 3.25 by 4-inch (82.6 by 101.6mm) slide having a uniform density of 2.0 or greater. The thermocouple shall be on the emulsion side away from the light source. The temperature rise in degrees above 70°F (21°C) ambient temperature shall be measured with an instrument accurate to ±3.6°F (2°C). The temperature rise shall be measured after the projector has been in continuous operation for a period of time to reach equilibrium and after the slide has been in projection position for at least 2 minutes. Nonconformance to 3.8.5.1 shall constitute failure of this test.

4.6.5.5.2 Temperature rise of projector. The projector shall be operated continuously for 8 hours at an ambient temperature of 70°F (21°C). The temperature rise at the hottest spot of the projector housing exclusive of the top cover of lamphouse and air exhaust port shall be determined by the thermocouple method after two hours of operation. At the end of the 8 hour period, the projector shall be examined. Nonconformance to 3.8.5.2 shall constitute failure of this test.

4.6.6 Environmental.

4.6.6.1 Low temperature. The projector shall be tested in accordance with MIL-STD-810, Method 502.1, procedure I. The storage temperature for step 2 shall be -60°F (-51°C). The operating temperature for step 4 shall be 32°F (0°C). Nonconformance to 3.9.1 shall constitute failure of this test.

4.6.6.2 High temperature. The projector shall be tested in accordance with MIL-STD-810, Method 501.1, procedure II. The operating temperature for step 7 shall be 110°F (43°C). The storage temperature shall be 160°F (71°C). Nonconformance to 3.9.1 shall constitute failure of this test.

4.6.6.3 Humidity (moisture-resistance). The projector shall be tested in accordance with MIL-STD-810, Method 507.1, procedure II, except that the number of cycles shall be 3. At the completion of the cycling, the projector shall be subjected to the tests specified in 4.6.1 and 4.6.7. Evidence of corrosion, breakdown of dielectric strength or degradation of operation shall constitute failure of this test.

4.6.6.4 Altitude (elevation). The projector shall be tested in accordance with MIL-STD-810, Method 500.1, procedure I. The operating altitude shall be 10,000 feet (3048m) and the nonoperating altitude shall be 50,000 feet (15,240m). Evidence of damage, erratic operation or malfunction of the projector shall constitute failure of this test.



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4.6.6.5 Drop. The carrying case, with the projector and accessories installed in it, shall be dropped twice upon a concrete floor from a height of 18 inches (457.2mm). At each drop, the base of the carrying case shall be parallel with the floor. The case shall then be opened and the equipment examined. Nonconformance to 3.9.4 shall constitute failure of this test.

4.6.6.6 Vibration. The projector in its carrying case shall be tested in accordance with MIL-STD-810, Method 514.2, procedure IX, part 2. Nonconformance to 3.9.5 shall constitute failure of this test.

4.6.6.7 Fungus. The projector and accessories (glass optical components excluded) shall be tested in accordance with MIL-STD-810, Method 508.2. Spraying shall include parts and materials within the interior of the equipment case. The equipment shall remain in an open condition during the test. Operation during testing is not required. Nonconformance to 3.9.6 shall constitue failure of this test.

4.6.7 Operation. The projector shall be set for operation and a slide shall be projected on the screen. The tilting device shall be operated and the focusing mechanism adjusted. Nonconformance to 3.10.1.1 or 3.10.2.3 shall constitute failure of this test.

4.6.8 Physical stability and off level operation. The projector shall be set up for operation and the tilting device fully extended. The supporting surface shall be inclined 20° in eight different orientations. Nonconformance to 3.10.1.2 shall constitue failure of this test.

4.6.9 Rigidity.

4.6.9.1 Slide changer. The projector shall be set up on a smooth masonite table or similar sturdy support surface. A cross hair image shall be projected onto a vertical screen located 20 feet (6.1m) from center of the projector. Nonconformance to 3.10.1.3.1 shall constitute failure of this test.

4.6.9.2 Ventilation motor. With the projector set up as specified in 4.6.9.1 and the tilt device extended to the maximum extension, the projected image shall be observed. Nonconformance to 3.10.1.3.2 shall constitute failure of this test.

4.6.9.3 Projector. With the projector set up as specified in 4.6.9.1, one end of the projector shall be lifted to a height of 4 inches (101.6mm) and dropped. The average time required for cessation of observable vibrations of the cross hair image shall be determined from at least six observations. Nonconformance to 3.10.1.3.3 shall constitute failure of this test.

4.6.10 Visual and mechanical inspection. The projector shall be examined for the defects listed in table IV.



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Table IV. Classification of visual and mechanical defects.

| Classification | Defect |
|---|---|
| Major | 101. Cleaning not as specified.<br>102. Fastenings not as specified or improper fit.<br>103. Projector does not operate with specified electrical input.<br>104. Overload protection not as specified.<br>105. Safety measures not as specified.<br>106. Construction of projector body not as specified.<br>107. Ventilation not as specified.<br>108. Switches not as specified.<br>109. Lens not as specified.<br>110. Lens mounting not as specified.<br>111. Bellows not as specified.<br>112. Condenser lenses not as specified.<br>113. Heat retarding filter missing or not as specified.<br>114. Optical parts not accessible for cleaning.<br>115. Built-in pointer not as specified.<br>116. Projection lamp not as specified.<br>117. Lamp and reflector adjustment not as specified.<br>118. Light shielding of lamp housing not as specified.<br>119. Power cable missing or not as specified.<br>120. Slide changer not as specified.<br>121. Carry case not as specified.<br>122. Weight not as specified.<br>123. Workmanship not as specified. |
| Minor | 201. Final paint film not as specified.<br>202. Identification marking incorrect, improper size, incomplete, or illegible. |

4.7 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First article inspection of packaging.
b. Quality conformance inspection of packaging.



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4.7.1 First article inspection of packaging. Unless otherwise specified in the contract, first article inspection of packaging shall be in accordance with the unit pack design validation requirements of MIL-P-116.

4.7.2 Quality conformance inspection of packaging.

4.7.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.7.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B quality conformance inspection requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.7.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in table V, "Packing inspection provisions". Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and acceptable quality level of 4.0 percent defective.

TABLE V. Packing inspection provisions.

| No. | Characteristic | Method of inspection |
|---|---|---|
| 101 | Intermediate container not as specified. | Visual |
| 102 | Improper closure of intermediate container. | Visual |
| 103 | Shipping containers not in accordance with specification. | Visual |
| 104 | Excessive cube. | Visual |
| 105 | Improper blocking and bracing. | Visual |
| 106 | Closure not in accordance with specification. | Visual |
| 107 | Weight and size exceed container limitations. | Weight & measure |
| 108 | Strapping not in accordance with specification, incorrectly applied, or omitted. | Visual |
| 109 | Marking omitted, incorrect or illegible. | Visual |

5. PACKING.

5.1 Preservation. Preservation shall be as specified in MIL-STD-726, coded as follows:

5.1.1 Level A. JR-1-B-ØØ-00-NS-X-ED-Ø-00-A.

5.1.2 Level B. JR-1-B-ØØ-00-NS-X-ED-Ø-00-B.



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5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

6.1 Intended use. Projector, Still Picture AP-4( ) is intended for use in field and fixed installations for projecting 3.25 by 4-inch (82.6 by 101.6mm) slides.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When a first article is required for inspection and approval (see 3.2).
c. Group C failure action required (see 4.5.4.2).
d. Level A or level B preservation and packing (see section 5).
e. When first article inspection rough handling tests are not required.
f. When first article inspection test reports require acquisition activity approval prior to production unit packing.

6.3 First article. When a first article inspection is required, the item will be tested and should be a preproduction sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for number of samples, examinations, test and approval of the first article.

6.4 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AP-4W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract (see 1.1).

6.5 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage and handling.

6.6 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

Custodians:
Army - ER
Navy - SH
Air Force - 99

Preparing activity:
Army - ER

(Project 6730-0116)

Review activity:
DLA-GS

User activities:
Army - MI
Navy - MC



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-P-15127D | PROJECTOR, STILL PICTURE AP-4( ) |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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